

Palestine Ahliya University
Faculty of Graduate Studies
Data Pre-Processing
Assignment 3

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Abstract

One of the biggest challenges in modern days is dealing with high-dimensional and noisy data. To handle these datasets, feature selection methods are utilized to reduce redundant features and save companies from wasted resources. Additionally, if the non-relevant features are dealt with, then the model can have better generalizability. Several factors play a role in determining which feature selection technique to use, such as data types, linearity of the model, and the size of the data. Alas, the best approach only becomes clear after experimentation.

Keywords: Feature Selection, Machine Learning, Filter Methods, Wrapper Methods, Intrinsic Methods, Chi-Squared Test, Mutual Information, ANOVA F-Test, RFE (Recursive Feature Elimination), Random Forest, CART, Permutation Importance, High-Dimensional Data, Model Accuracy, Dimensionality Reduction, Supervised Learning, Numerical Features, Categorical Features, Model Generalization.

Introduction

In this research paper, we will employ various methods to apply feature selection techniques that are crucial for enhancing model performance and accelerating model training. We will be using three datasets: one that has a numeric input and a categorical output, one with a categorical input and a categorical output, and one with both inputs and outputs being numerical. It is essential to explore many different approaches to finding the best techniques for each data type.

There are two types of feature selection, which are supervised feature selection and unsupervised feature selection. In this paper, we will focus on supervised learning using three supervised feature selection methods: Filter, Wrapper, and Intrinsic methods.

Methodology

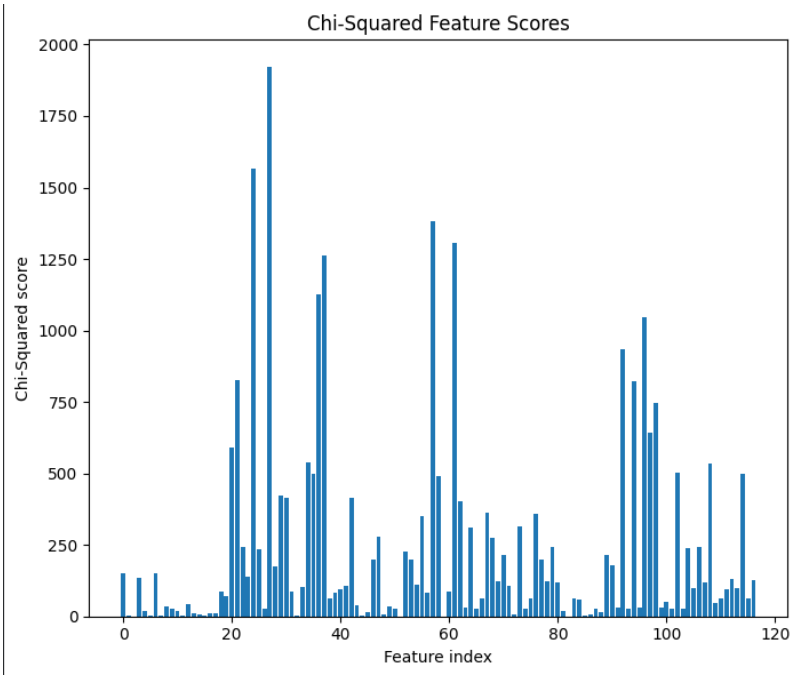
The filter method is a technique that is used to point out the dominant features in a dataset. First, we rank the features by their scores using different methods, and then select the k-top inputs that have the highest scores as our best features. Furthermore, we drop the features that do not have a good score because they only add noise to our model, potentially lowering the model's accuracy. Using the filter method, we will be analyzing the results from two datasets: waterQuality1.csv (numerical inputs), which explains the effects of different minerals and chemicals found in water, and the Mushroom dataset (categorical inputs) for recognizing which mushrooms are edible or poisonous for human consumption. For the Mushroom dataset, we will be using the filtering methods Chi-squared and Mutual Information statistics.

Chi-Squared statistic determines if there are meaningful relationships between the input variables and the output variables. It does this by finding the features that are independent of the target variable and removing them completely. Independent features from the target variable only add noise and affect our model's accuracy. Applying this method finds the best features using statistics by selecting the top k-most relevant features in the dataset.

```
fs_chi2 = SelectKBest(score_func=chi2, k=k)
```

Figure 1: selecting the top *k* most relevant features

After finding out which K to use, we split the data into training and testing to find the most irrelevant features in our dataset. The results of applying the Chi-Squared feature selection in the dataset are as follows:



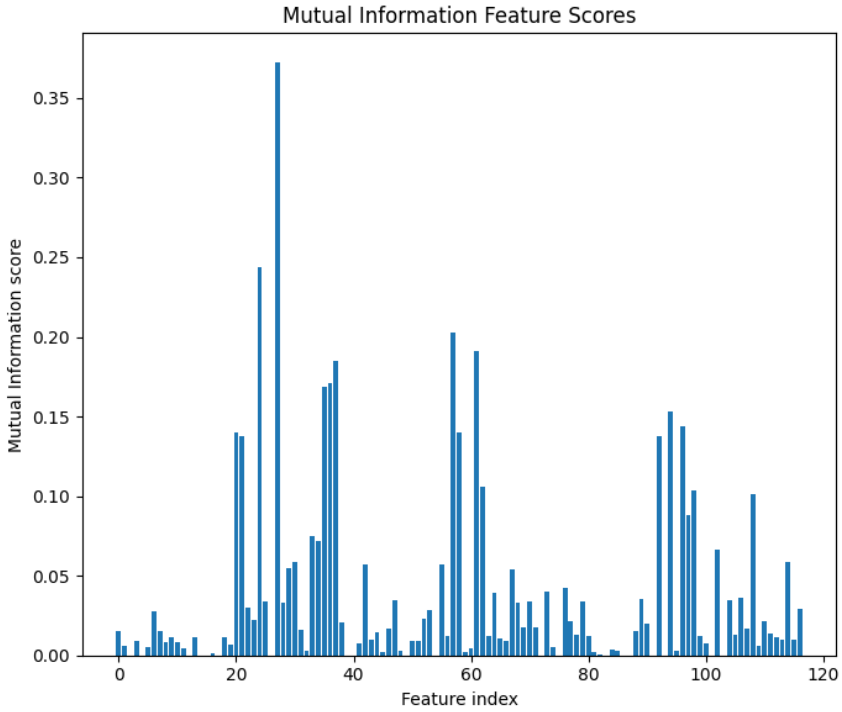

```
Total features after one-hot encoding: 117

Chi-Squared Selected Features Accuracy: 97.58%
Top Chi-Squared features (indices): [21, 24, 27, 36, 37, 57, 61, 92, 94, 96]
```

Figure 2: The results and accuracy using the Chi-Squared statistic

First, we used the one-hot encoding method to convert some categorical values into numerical values to improve the performance, and it dropped 3 features, indicating that there was a multicollinearity problem with the model that the one-hot encoding method fixed. Our model's accuracy is exceedingly high at 97.58% (See Figure 2), indicating that the Chi-Squared method removed redundant features and kept the most important ones that have a high correlation with the target variable (is_edible).

Another method for categorical feature selection we will be exploring for this dataset is the Mutual Information. It calculates the scores of each input variable and makes a comparison of each input feature to the target variable. Features are then selected based on this score; if a feature scores high, then it is used for the model. On the other hand, if it scores lower than a certain threshold, then it is discarded.




```
Mutual Information Selected Features Accuracy: 97.58%
Top Mutual Information features (indices): [20, 24, 27, 35, 36, 37, 57, 61, 94, 96]
```

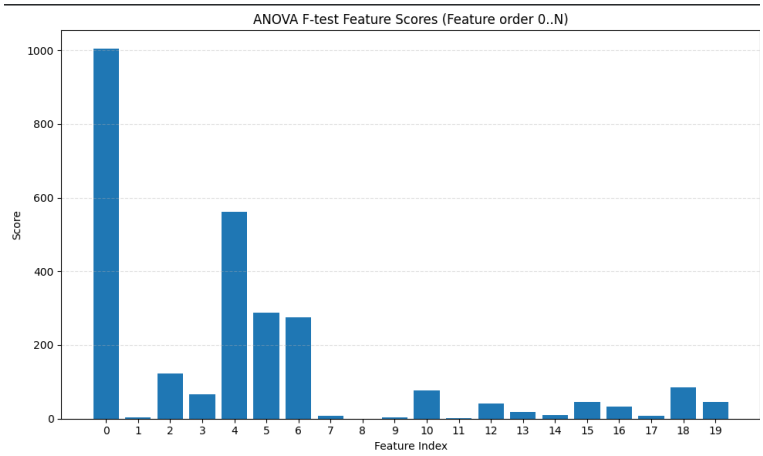
Figure 3: The result and accuracy using Mutual Information

If we compare both feature selection techniques, we can see that although they have both removed different features, they still have the same accuracy (97.58%)

(See Figures 2 and 3). This is likely due to both metrics having a strong mathematical relationship and both ranking the features based on the dependency of the target variable.

For our other dataset (waterQuality1.csv), which has numerical inputs, we will be using other filtering methods such as the ANOVA test and mutual information. Since the Mutual Information method works on both categorical and numerical data, we will be implementing it alongside the ANOVA test and comparing their accuracies. The ANOVA feature selection method is used for calculating variance values for a group of samples (three or more) and computing whether they come from the same

distribution or not. If they do, then there is a relationship with the target variables, and conversely, if they do not, they would not have a relationship with the target variable. For this particular dataset, there were some preprocessing techniques, such as dropping rows, due to having missing values. After the preprocessing was complete on the water dataset, the ANOVA test was applied to find which features to keep. The results were as follows:



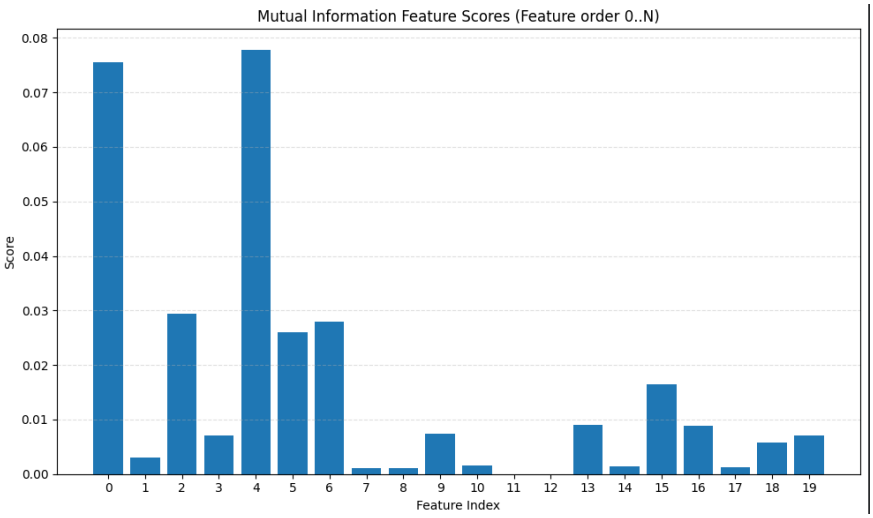

```
Accuracy (All features): 89.81%

================ ANOVA RESULTS ================
Best K (ANOVA): 11
Best Accuracy (ANOVA): 89.84%
Selected ANOVA feature indices: [0, 2, 3, 4, 5, 6, 10, 12, 15, 16, 18]
ANOVA removed 9 features
```

Figure 4: The result of using ANOVA on the water dataset (using the best K)

As we can see from the results in Figure 4, knowing the percentage of Aluminum, Barium, Cadmium, and Chloramine in the water is detrimental to finding out whether the water is safe or not. These are the top features, along with some others, that have the strongest relationship with the target variable. We can see that ANOVA removed nine features and improved the accuracy a little. Although the accuracy did not significantly improve, it still reduced noise, reduced computational cost by removing the features, and potentially improved generalization.

Utilizing Mutual Information on the same dataset results in a different outcome. Some features that were chosen to have the strongest relationship with the output variable were distinct from the ones chosen in the ANOVA F-Test. Below are the results and accuracy of the model using Mutual Information as a numerical feature selection method.

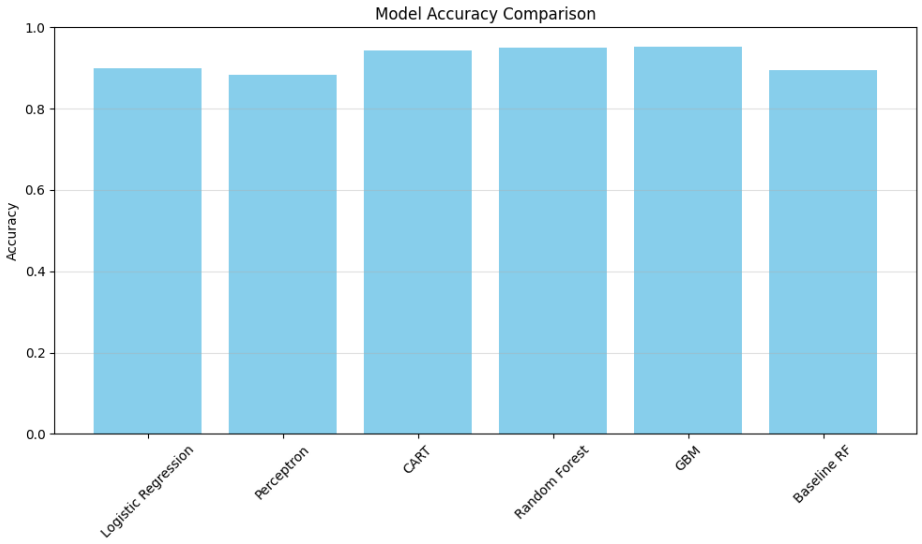



```
================ MI RESULTS ================
Best K (MI): 11
Best Accuracy (MI): 89.96%
Selected MI feature indices: [0, 2, 3, 4, 5, 6, 13, 14, 15, 16, 18]
MI removed 9 features
```

Figure 5: The result and accuracy of using Mutual Information in a numeric dataset

Comparing Figures 4 and 5, we can see that Mutual Information selected different redundant features compared to ANOVA. The accuracy overall was very similar between both techniques, although Mutual Information had a slightly better accuracy. This is likely because ANOVA is only capable of measuring the linear relationship between the variables, whereas Mutual Information can capture non-linear relationships between the input and output.

The second type of supervised feature selection is the wrapper method. Wrappers' methods generally fit a model, evaluate features based on their importance, remove redundant features, and refit the model with fewer features. This is done iteratively until we can find the K-best number of features with the best possible accuracy for the model. Below is a model embedded with an RFE (Recursive Feature Elimination) wrapper that improves the model's accuracy depending on different algorithms:




```
========== FINAL ACCURACY SUMMARY ==========
Logistic Regression   : 90.00%
Perceptron            : 88.38%
CART                  : 94.25%
Random Forest         : 95.00%
GBM                   : 95.25%
Baseline RF           : 89.44%
```

Figure 6: Using RFE with different algorithms

The RFE works on both numerical and categorical data. In this case, we used the Mushroom categorical dataset. We can see in Figure 6 that the best accuracy is achieved using RFE with the GBM algorithm. This is because it is a powerful method that combines weaker methods, such as decision trees, and trains the data in a sequential

order, leaving minimal errors in the process. RFE uses intrinsic methods in its core and other techniques, like Lasso, decision trees, random forest, and GBM, to iteratively remove features and find the least number of features with the highest accuracy possible.

Another method for feature selection is permutation feature selection. It can be used as a wrapper feature selection method. This method permutes feature values and discovers the changes in the model's accuracy depending on the sorting of these features together. Below is an example of classification permutation on the water dataset.

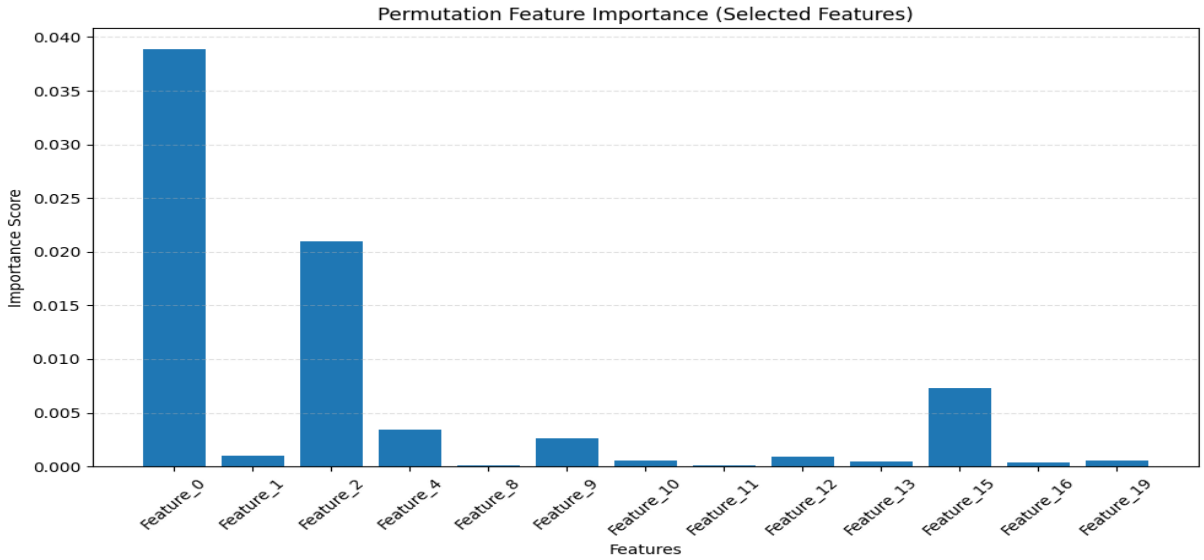
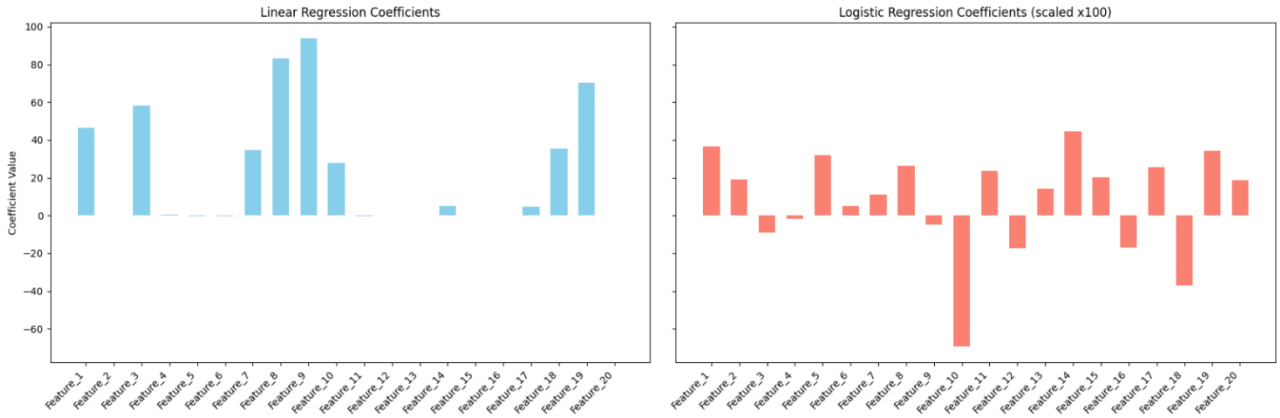


Figure 7: Using Permutation to find the important and irrelevant features

As we can see from Figure 7, the model had an accuracy of 89.81%, which was better than the ANOVA and Mutual Information tests. This is likely because ANOVA fails to deal well with skewed data, and Mutual Information often struggles with small sample data, such as the chosen dataset in this research paper.

Lastly, some techniques help identify which features are relevant for the model's prediction, such as Feature Importance. It helps observe patterns and model behavior by ranking features based on their usefulness to the target variable. It can give apprehension into which features should be kept or removed. This synthetic dataset, along with the results, will give insight into how Feature Importance can help remove redundant features and determine which features have the highest correlation with the target variable.



	Feature	LinearRegression	LogisticRegression
0	Feature_1	46.325998	0.365327
1	Feature_2	-0.034162	0.192593
2	Feature_3	58.330507	-0.092085
3	Feature_4	0.428824	-0.018583
4	Feature_5	-0.429282	0.318828
5	Feature_6	-0.371167	0.051909
6	Feature_7	34.541479	0.110655
7	Feature_8	83.323917	0.263697
8	Feature_9	93.923940	-0.047535
9	Feature_10	27.815506	-0.694468
10	Feature_11	-0.122906	0.238142
11	Feature_12	-0.043569	-0.175397
12	Feature_13	-0.057731	0.140888
13	Feature_14	4.891230	0.445823
14	Feature_15	0.245706	0.202846
15	Feature_16	0.085308	-0.171273
16	Feature_17	4.833278	0.255383
17	Feature_18	35.585044	-0.370857
18	Feature_19	70.479709	0.340979
19	Feature_20	-0.093245	0.187754

Figure 8: Using 2 synthetic datasets to show how feature importance operates

We used a linear regression model (numerical data) and a logistic regression model (categorical data) using two different synthetic datasets. The feature number and the sample size are the same. As seen in Figure 8, we can see which features are relevant according to their values, and which features we should remove, due to their low value.

Lastly, we select the subset of the most important features and delete the rest to reduce dimensionality and increase accuracy. (Negative numbers do not represent bad relationships; they just represent the direction of the relationship).

Result

As the complexity of data is always increasing, feature selection techniques are required to reduce redundant features. We tested different techniques on different types of data. Mutual Information and RFE (Recursive Feature Elimination) worked on both numerical and categorical data, while different techniques, such as ANOVA and statistical feature selection, such as Chi-Squared, worked on only one type of data. We explored different methods, in particular, filter, wrapper, and intrinsic methods for supervised feature selection. Although the wrapper methods gave the best accuracies, the run time was really slow, and for big datasets, this would be computationally expensive and memory-intensive to run. For simple datasets with no high-dimensionality, a statistical-based method, such as the Chi-Squared statistic, would give good results, and a fast runtime for the model to run. Accuracy is not always the only vital aspect for a good model, but also the generalization on new data and runtime, which is hard to do without a good feature selection technique.

Conclusion

The process of feature selection is vital for building a high-quality model. It is essential to remove noise and unnecessary features that negatively impact the model's accuracy and cause overfitting and increased runtime. In this research paper, we explored different techniques for feature selection, namely supervised ones such as Intrinsic, Wrapper, and filter methods. Filter methods gave good results and fast runtimes, but the issue with filter models is that they are independent of the model. This means that sometimes, the chosen filter model does not work as intended and selects redundant

features instead of relevant ones. This is evident by comparing the models with the filter methods and the models with the wrapper and intrinsic methods. Although the wrapper techniques had better accuracy, their runtime takes too long, especially for a high-dimensional dataset. In the end, no one knows which feature selection method works best. Yes, we can use specific techniques for certain data types, but trial-and-error is the only way to know which method performs well for a specific dataset.

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